



EMIA6500R · CMAA5022 · CSM160021

# Social Media for Creatives

## Lecture 05

Homophily, Creative Advertisement  
and Viral Marketing

Professor Pan HUI; FRI 15:00 – 18:00

# Today's Outline



01

## Lecturing

Homophily, Creative Advertisement and Viral Marketing

02

## AI Lecturers

Neil and Jennie

03

## Lecturing

Homophily, Creative Advertisement and Viral Marketing

04

## Paper Reading and Presentation

Group 1: Social Media for Academia

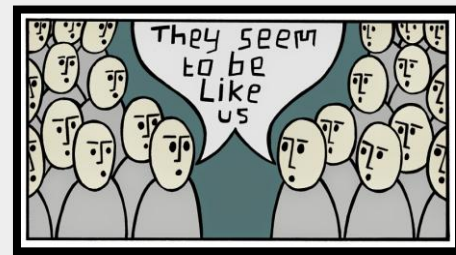
05

## Participation and Group Discussion

Group Task 2

# Background: Similarity breeds Connection

- People with **distinctive characteristics** (e.g., genders, races, ages, etc.) appear to have **quite different similarities**.
- This may be **bias** but we often **attribute these similarities** to some essential aspects of their category membership. *For example, women are emotional, educated people are tolerant, and gang members are violent.*
- People usually have more significant contact with others like themselves, suggesting that these similarities tend to become localized in social relations. It comes to **typify “people like us” →**



**Homophily!**

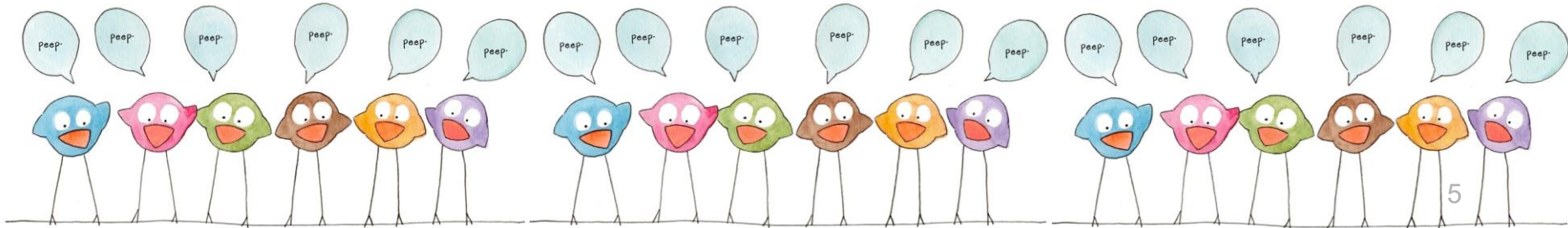
# What is Homophily?

- In our social relations, homophily is the **principle that contact between similar people occurs at a higher rate than among dissimilar people.**
- As in the proverb “birds of a feather flock together” and 物以类聚，人以群分 – 《战国策·齐策三》
- In social networks, **network homophily** refers to the fact that, based on node attributes, similar nodes may be more likely to link to each other than dissimilar ones.
- Some scientists also interpret homophily as **network distance** that could be translated in terms of social characteristics.

# Type of Homophily - 1

Lazarsfeld & Merton (1954) first distinguished two types of homophily based on **social status characteristics**

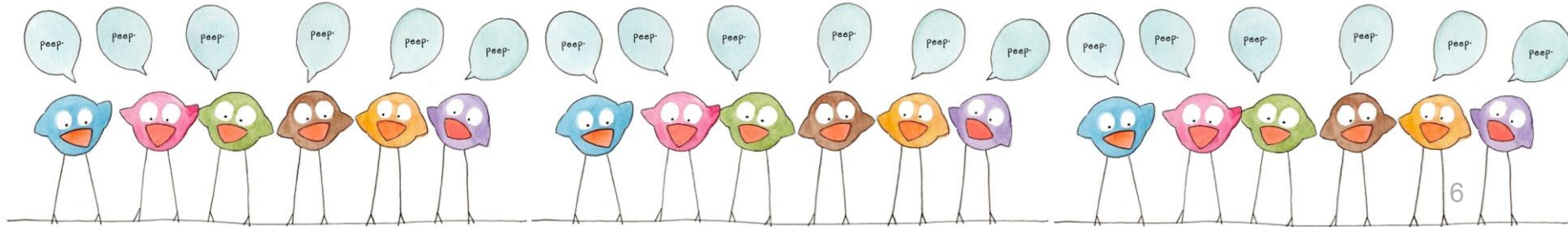
- **Status homophily** refers to the tendency to form social connections with others with **similar informal, formal, or ascribed status**.
  - Including both **assigned-at-birth characteristics** (e.g., race, sex, and age) and **acquired characteristics** (e.g., religion, occupation, and education)
- **Value homophily** refers to the tendency to form social connections with others with **similar values, attitudes, and beliefs**, regardless of differences in status characteristics.
  - Including **the wide variety of internal states** presumed to shape our orientation toward future behavior



# Type of Homophily - 2

Later researchers expended the categories by considering diverse distributions on population and personal preferences

- **Baseline homophily** refers to **tendency to form social connections that exists even in an uneven distribution of people with varying characteristics.**
  - E.g., People tend to choose others with same genders when it comes to connection based on social comparison (Miller, 1984)
- **Inbreeding homophily** refers to **the similarity typically due to personal preferences and choices.**
  - E.g., Marsden revealed that teenagers tend to befriend and confide in others of similar ages (Marsden, 1988)



# Causes and Examples of Homophily

## Geography

People **located nearby have similar characteristics** and are more likely to form contact with each other. Technology such as the telephone, e-mail, and social networks have **reduced but not eliminated** this effect.

## Family Ties

Family relationships are generally **close and keep frequent contact** though they may be at great geographic distances.



## Organizations

**School, work, and volunteer activities** provide the great majority of non-family ties.

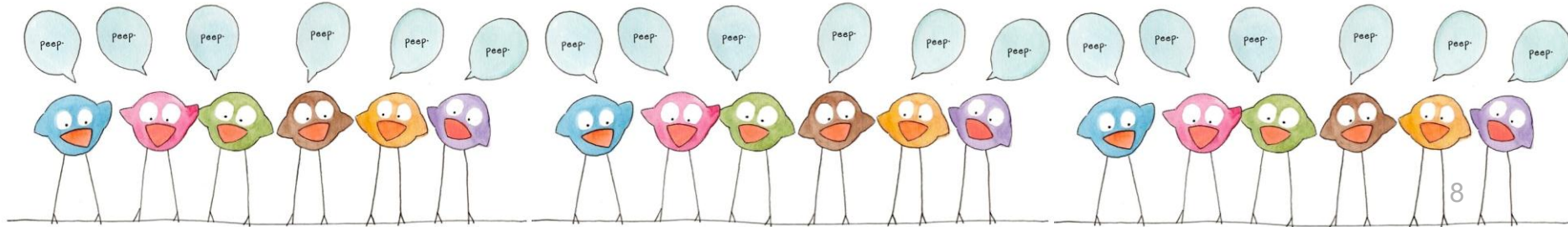
## Cognitive Processes

People who have **cultural similarity** tend to own shared knowledge, and more easily to form communication and share cultural tastes.

# Type of Homophily - 2

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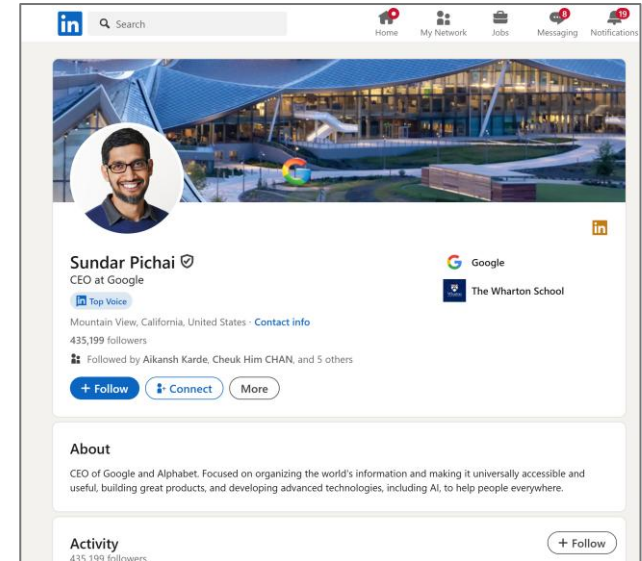
# Homophily on Social Media

- Features and functions on social media also **breed new relations (ties) between users**, *e.g., likes, following, commenting*. Homophily can also be observed through these new relations.
- Homophily can be seen in the way users **form connections and interact** with others on social media platforms.
- For example, *people with similar interests are more likely to follow and interact with each other on platforms like X, TikTok, Instagram, and Threads.*
- Similarly, people from **the same geographical region or social background** may be more likely to connect on social media platforms. *i.e., Chinese use Weibo and Russian use VKontakte.*



# Homophily on **User's Profiles**

- Users' profile can **reflect homophily** on social media.
- A user profile is a page that represents a specific user on a social media platform.
- Users can present themselves on profiles that reflects their **social characteristics and interests**, which can lead to the formation of homophilic networks.
- For example, on LinkedIn, users can list their education and professional experience, which can be used to connect with others in the same industry or field.



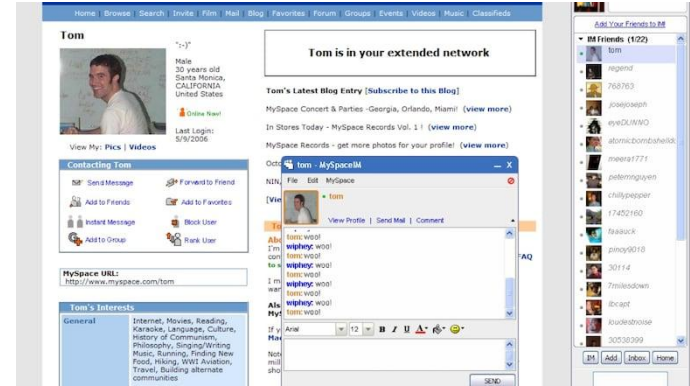
# MySpace – Profiles Foster Homophily -1

- Some researchers have explored the attributes on social media profiles that could enhance homophily on MySpace.
- MySpace is a U.S. social network site (SNS) launched in 2003, where users can form relationship as Friend.
- Myspace user has **a profile page with self description** (e.g., a selfie photo, status, and list of interests) and **list of Friends that is visible to Friends or other strangers** (if set as not private).
- It has been shown that MySpace users' profiles seem to be used often for **identity expression**, which could also influence, or even determine, users' choice to become Friends (Boyd, HICSS 2016).



# MySpace – Profiles Foster Homophily -2

- It has been widely recognized that MySpace profiles could **foster Friend homophily**.
- For instance, one previous study has analyzed an aspect of homophily in a large-scale study of *MySpace Friends: the tendency for Friends* to have similar interests or tastes on profiles.
- Mike then took a closer look on **what attributes on users' profiles preset significant similarities among Friends** (Mike, 2009).
- By investigating pairs of active MySpace Friends based upon a systematic sample of 2,567 users, the author showed **highly significant evidence of homophily on factors as age, country, marriage, etc.**
- His findings demonstrated how users' profiles fostered homophily among MySpace Friends, as well as uncovering crucial aspects significantly acting on such homophily.



# Online Dating – Success on Homophily -1



- On success story of apply homophily is online dating services.
- Online dating refers to **the method via the internet used by people to search for and interact with potential romantic partners.**
- In the context of homophily, online dating can be seen as a way to seek out and connect with romantic partners with **similar characteristics**, such as *age, education, income, interests, and values*.
- Homophily acts on the **matching algorithms** in online dating services.
- Many online dating platforms and apps to match users based on shared characteristics, preferences, and behaviors.
- This approach can lead to **successful matches based on homophily**, as users are more likely to connect with individuals who share similar backgrounds and experiences.

# Online Dating – Success on Homophily -2



- The success of homophily applied in online dating has been supported by lots of evidences from markets, industries and academics.
- According to marketing statistics, in 2022, there were **over 366 million online dating service users**. Implied by yearly increases rate, by 2027, it is estimated that there will be 440 million people seeking love through online platforms.
- Additionally, internet dating generated 2.86 billion U.S. dollars in 2022, with the market showing no signs of slowing down (Statista).
- The power of homophily are also reflected on breeding and promoting offline connections among users.
- It is reported that **76% of women and 83% of men** have met in person with someone they first talked to online (Cloudwars).

# Online Dating – Success on Homophily -3



- Researchers have put much effort into characterizing such success through users' **demographic features** and **behavioral patterns**.
- Fiore and Donath's study on demographic features uncovered **the strong effect of sameness on daily habits** (e.g., smoking, rearing pets, etc.) and **appearance** (e.g., eye colors, body buildings, etc.) in determining users' choices of partners to date (Fiore, CHI EA 2005).
  - Users were most strongly seeking dating partners same as themselves on these two major aspects.
- Gareth et al. explored Tinder users' behavioral patterns when viewing others' profiles and their survey showed that users across all genders tended to spend more time and attention on bios and profile pictures.
  - This is something as baseline homophily that can help Tinder users improve their online matching rates (Gareth, ASONAM 2016).

# Online Dating – Be Careful of Catfishing



- In the context of online dating, homophily can sometimes play a role in the occurrence of catfishing.
- Catfishing refers to the act of creating a false online persona with the intention of deceiving someone into a romantic relationship.
- Catfishing profiles can create a sense of trust and familiarity between individuals who share similar characteristics, where victims may feel it authentic and believable when seeing familiar information to themselves.
- Catfishing is deceptive and indistinguishable. Most people cannot identify their partners are catfishers even after they get matched and interact with each other for already several times.
  - In a survey on catfished users in U.S., only 31% of the participants identified that they had interacted with catfishers ([Statista](#)).

# Homophily Goes Wrong: Hate Speech - 1



- When it comes to hate speech, homophily can exacerbate its spread and impact.
- Hate speech refers to any form of expression that seeks to offend or insult individuals based on their identities, such as race, gender, sexual, etc.
- When people associate primarily with others who share same views and beliefs, they are less likely to encounter opposing viewpoints and perspectives (high homophily in communication), which can lead to echo chambers.
- In these echo chambers, hate speech can become normalized, as people are less likely to encounter counterarguments and perspectives that challenge hateful or intolerant views.

# Homophily Goes Wrong: Hate Speech - 2



- There are lots of evidences supporting a close relation between **homophily** and **spread of hate speeches**.
- The popularity of hate speeches in online communities can be bred by **different types of homophily**.
  - For instance, individuals speaking **common languages** are more likely to adopt hateful topics and spread them to each others (Chopra, AAAI 2020).
- In addition, homophily is also **observed among hateful users**.
  - A dataset study related to COVID-19 finds that **anti-Asian spreaders most are following same hateful influencers** and further form interconnections by spreading **same hateful topics from identical influencers** (He, ASONAM 2020).

# Today's Outline

02

## AI Lecturers

Neil and Jennie: Advertisement in Social Media



**NEIL**

- Gender: Male
- Age: 21
- AI VTuber and Social Media Influencer
- Digital content and AI and creativity



**JENNIE**

- Gender: Female
- Age: 24
- Nationality: unknown (mixed)
- Influencer and KOL with many followers

Available on our website



# Advertisement and Social Media

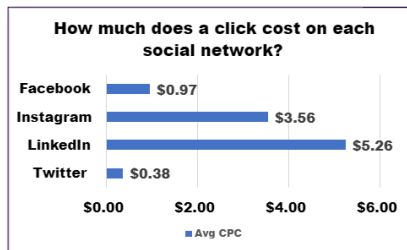
- Next, we explore **the advertisement on social networks** and examine some **examples** of its impact on marketing applications and social effects.
- Advertisements (Ads) are the practice and techniques employed to bring attention to a product or service.
- Social media provide new platforms for Ads to **promote products and services to audiences in virtual communities throw the Internet**, which are categorized as online advertisements.
- Nowadays, social media Ads have become a **massive industries** on many platforms, including Facebook, Instagram, Red (Xiaohongshu), etc.
- It is reported that, in 2022, social media Ads spend stood at USD 230+ billion. **Ads spending in the Social Media Advertising market worldwide is projected to reach USD 276.72 billion in 2025.**(Statista).





# What is Social Media Ads ? - 1

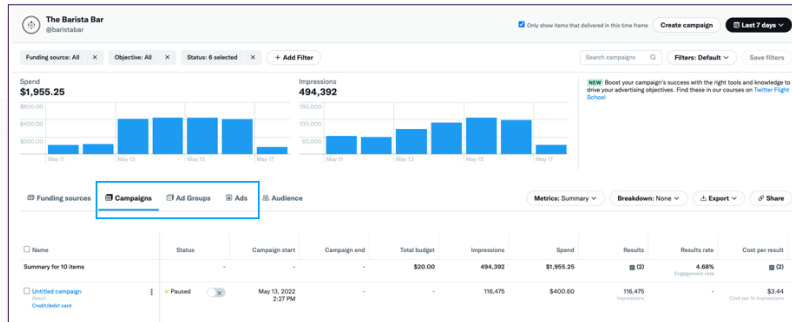
- There are many advantages to sell Ads on social media platforms.
- **Wide Reach:** Social media platforms have a large user base, and businesses can target specific segments of this audience based on factors such as age, gender, location, interests, and behaviors.
  - This allows businesses to create targeted and personalized ads that are more likely to **resonate** with their target audience.
- **Cost-effectiveness:** Compared to offline Ads, social media Ads are often much more **affordable**. Businesses only pay when someone clicks on the ad or takes a specific action, such as making a purchase.



*For example, the right figure shows several mainstream social media's cost-per-click (CPC) and businesses only need to spend several dollars then can advertise to a user! ([Source](#))*

# What is Social Media Ads ? - 2

- **Real-Time Performance Data:** Social media advertising allows businesses to track the performance of their ads in real-time.
  - i.e., *how many people have seen their ad, how many have clicked on it, and how many have taken the desired action, such as making a purchase.*
  - This data can help businesses make **informed decisions** about their advertising strategy and **adjust their campaigns** as needed to improve their performance.



*This is Twitter Ads Manager, a powerful central workspace to show real-time performance data. It can also generate statistics results and provide intelligent suggestion for Ads strategies.*



# Social Media Ads Go **Wrong** ... - 1

- However, advertising on social media can sometimes go wrong. Here are some typical type of misused Ads...
- **Deceptive Ads** refer to ads that use misleading or false information to promote a product or service.
  - These ads are designed to trick users into **believing something that is not true**, in order to encourage them to make a purchase or take some other action.
- For example, a company called **Roca Labs** paid social media influencers to post fake reviews of its weight-loss products. In the meanwhile, they also created fake user profiles to post positive comments.
  - In 2018, Roca Labs case was settled by Federal Trade Commission (FTC) and the company needed to pay \$500,000 for deceptive advertising activities ([Source](#)).



# Social Media Ads Go Wrong ... - 2

- People dislike **web ads** is that they are annoying and disruptive.
- These types of ads can take many forms, including **pop-ups**, **auto-play videos**, and **ads that take up the entire screen**.
- **Annoying and disruptive Ads** can cause negative effect on users' experiences and their attitudes to the markets.
  - It has been reported these ads can have a negative impact on users' experience, by making it difficult to navigate the site, consume content, and interact with other users (Brajnik, 2010).
  - Researchers also reveal the issues of these Ads on users' attitudes, like increased cognitive load, feelings of irritation among users, and reduced trust in the hosting websites (Burke, 2005).





# Social Media Ads Go Wrong ... - 3

- **Clickbait Ads** are very common misused Ads on social media platforms.
- **Clickbait** is a text or a thumbnail link that is designed to attract attention and to entice users to follow that link and read or view the linked content.
- **Clickbait Ads** usually utilize typically exaggerated, sensationalized, or otherwise misleading headlines to attract users' attention.
- Moreover, Clickbait Ads can be used for **propaganda purposes in news domains**, with targets as spreading misleading or deceptive claims, mis/disinformation, and voter suppression.

*This is a screenshot of the site called Mommy Underground sharing anti-vaxx content using "Shocking" clickbait headline.*

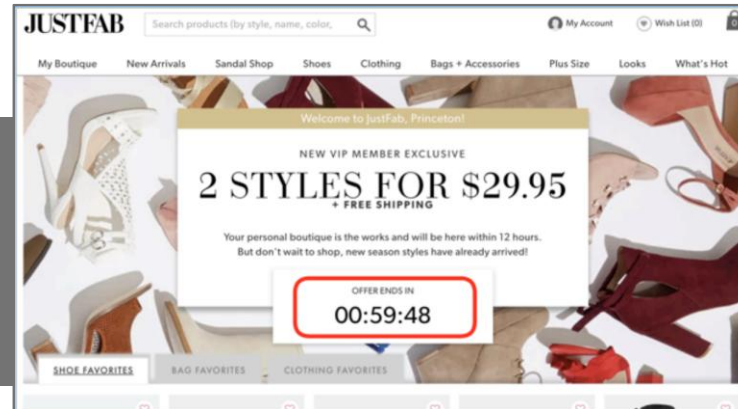


# Be Aware of Dark Patterns



- **Dark patterns** are user interfaces that benefit an online service by leading users into making decisions they might not otherwise make.
- Dark patterns can deceive users by **covertly manipulating or coercing them into choices that are not in their best interests**.
- In the context of advertising, dark patterns are used to **coerce or manipulate** users into clicking on ads, making purchases, or providing personal information.

*This shows a widely used type of dark patterns – countdown timer. Such patterns can cause an implication of stress for losing the sales, pushing users to purchase the products.*



# What makes a “Bad” Ad? - 1

- So, what are **the factors we should take care of to avoid our Ads being blocked?**
- Motivated by these questions, Zeng et al. explore the knowledge of **which types of Ad content users consider problematic and detrimental** to their browsing experience (Zeng, CHI 2021).
- By performing a survey on U.S. participants (N=60) who have experienced consuming social media Ads, the authors create **a taxonomy of 15 positive and negative user reactions to Ads' contents**.
  - This taxonomy has also been verified by supporting statistics results on a labeled Ads dataset (N=50) by crowdsource (N=1000).
- These findings suggest crucial factors for Ad designers to **improve content quality on users' experience**. Next, we will introduce several inspiring factors.

# What makes a “Bad” Ad? - 2

- Here are several implications based on participants’ response that would make social media Ads feel bad.
- **Clickbait elements**
  - Participants feel that these attention-grabbing techniques are “condescending”.
  - In addition, participants experience large frustration when observing how clickbait Ads tend to omit or conceal information in the Ad, to bait them into clicking it.
  - *“I hate any of the ads that say things like “You won’t believe what...” or “They’re trying to ban this video...” or nonsensical click-bait hyperbole.”*
- Psychologically manipulation: Participants dislike ads that are **trying to manipulate their emotions and actions**, such as ads make them feel unwanted emotions, especially “fear” or “panic”.



# What makes a “Bad” Ad? - 3

- **Distasteful, offensive, and uncomfortable content**

- Participants reacted as “repulsed” or “gross” to some ads with disgusting, offensive or sexually suggestive content.
- Such as *an Ad with a poorly lit picture of sprouting chickpeas in front of a banana.*



- **Cheap, ugly, and low-quality Ads**

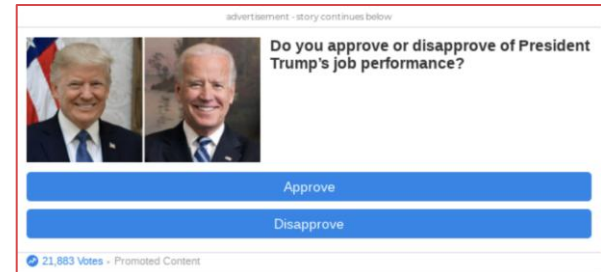
- The **quality of aesthetics** in Ads also acts on participants' experience.
- It is reported that the poor quality of the Ad reflect poorly on the product, saying that it makes the company look “desperate”, or that it made them think the Ad looked like a scam.
- Like *small fonts or “garish”, too-bright colors.*



# What makes a “Bad” Ad? - 4

- **Political content**

- Participants **disliked politics** in their ads, for different reasons, such as **disagreement on political ideologies** or that the **participants just don't expect any political message or tones** in an advertisement.
- *“I dislike everything. At least there's no stupid President in my face, but come on, get your politics and agenda away from me. I even agree with this ad but it's still managing to annoy me! Go away!”*
- Participants also feel **uncomfortable** for an Ad that looks like a **political poll** intending to activate their political beliefs and lure them into clicking to support their preferred candidate



# What makes a “Good” Ad ? - 1

- Similar, we will discuss about implications that would make social media Ads feel good.
- **Trustworthy and genuine content**
  - Participants responded positively to ads that they described as *“honest”, “trustworthy”, “legitimate”, and “authentic”*.
  - Some signals people cited for these traits include ads that look “refined” and high quality, images that accurately depict the product, and ads that include brands they recognized.
- **Good design and style**
  - Participants like *aesthetically pleasing* ads.
  - Like pleasing color choices, images that are “eye-catching”, interesting, beautiful, or amusing, and a “modern” design style.

# What makes a “Good” Ad ? - 2

- **Entertaining and Engaging Ads**

- Participants like ads that they find **entertaining, engaging**, or otherwise gave them **positive** feelings.

- **Relevant, interested in the product**

- Many participants, when asked about what kinds of Ads they like in general, say that they enjoy Ads which are targeted at **their specific interests**.
- Various participants mention liking ads for their specific hobbies, food, pets, and for products they are currently shopping for, etc.

- **Simple and Straightforward**

- Participants appreciate Ads that are **easy to understand, and straightforward** about what they were selling.
- *“When I am browsing, I enjoy ads that are unique and advertise the brand name clearly, without disrupting the content I am viewing. Specifically, side banners and top banners are fine.”*

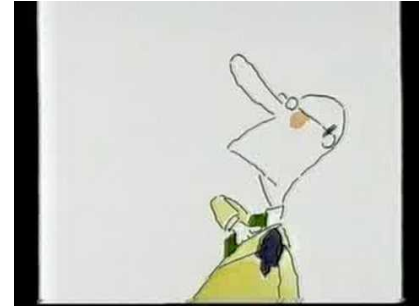
# Social Media Ads in Text Domain



- A Social Media Ads can contain **multiple multimedia elements**, like **text** for introduction, **images** showing products and **videos** presenting product usages.
- Next, we zoom into two crucial domains – **text and video** – and discuss about important elements and their applications in social media advertising.
- First is **the slogan of the Social Media Ads**.
  - According to Oxford dictionary, a slogan is *“a short and striking or memorable phrase used in advertising.”*
  - In the context of social media advertising, slogans can be an effective tool for **grabbing users' attention and conveying a brand's message** in a concise and memorable way.

# Successful Slogan Ads: #GivesYouWings

- A successful slogan can significantly boost revenue for businesses throw social media platforms.
- One example is the success of Red Bull.
  - #GivesYouWings is a hashtag campaign created by Red Bull, a popular energy drink company. This campaign is an extension of their well-known slogan "Red Bull gives you wings".
  - This campaign is designed to create a sense of community around the Red Bull brand, and to encourage customers to share their positive experiences with others.
  - As an online advertising strategy, #GivesYouWings allows Red Bull brands to reach a wider audience than traditional advertising channels.
  - It is reported that there has been more than 620,000 followers to #GivesYouWings on X.
  - On YouTube, #GivesYouWings videos have generated more than 284 million views (Source).



Red Bull Gives You Wings 2000



Red Bull Gives  
You Wings  
YouTube  
Shorts 2024

# Why Slogans are Important? -1

- Without doubt, **Red Bull's slogan is an impressive success** in advertising and helps brands to broaden marketplace on social media.
- But why slogans are important? What factors make them so powerful in advertising?
- The answer is **brand identity**.
- Brand identity refers to **the visible and tangible elements of a brand** that represent the company's values, personality, and mission to its customers and the public.
- Together with brand name and logo, the slogan represents one of the **three key elements of brand identity**; that is, the elements by which the brand communicates with all their customers (Bradley, 2002).



If you fall, fall in Dior

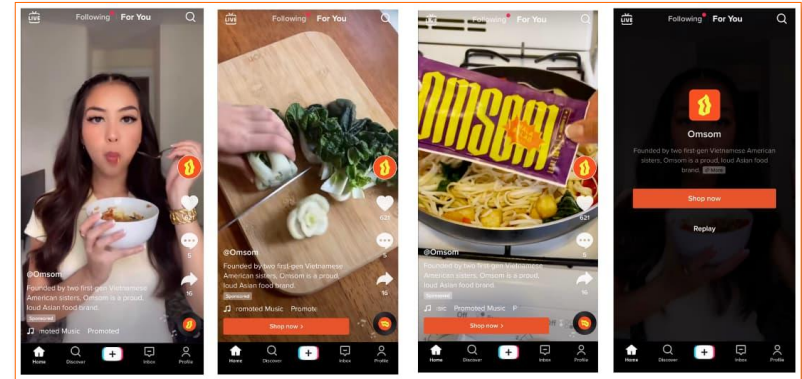
## Why **Slogans** are Important? -2

- Slogans also act on a primary aspects that influence brand identify – **brand awareness**.
- Brand awareness refers to customers' ability to **remember the brand** without any (or with a little) aid.
  - A well-crafted slogan can help a brand to stand out in a crowded market, create a lasting impression and long-term memory in the minds of consumers.
  - For example, the iconic slogan “**Just Do It**” has been synonymous with Nike for over 30 years.



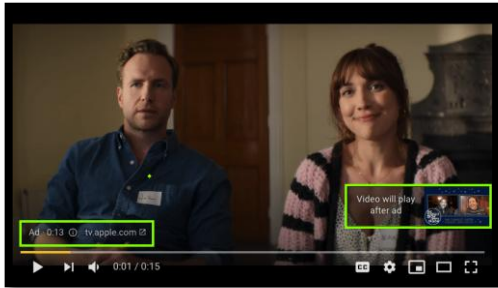
# Social Media Ads in Video Domain -1

- **Video Ads** are a major component of social media Ads.
- Compared with text and image Ads, video ads offer a **unique opportunity for brands to tell a story**, showcase their products or services, and connect with their audience on an emotional level (Wendt, 2016).
- For many social video platforms, video Ads have become one of the mainstream revenue resources.
- For example, in 2022, TikTok's Ad revenue has exceeded 11 billion U.S. dollar, suggesting TikTok has earned 65.8 dollar from each of its users by selling video Ads.

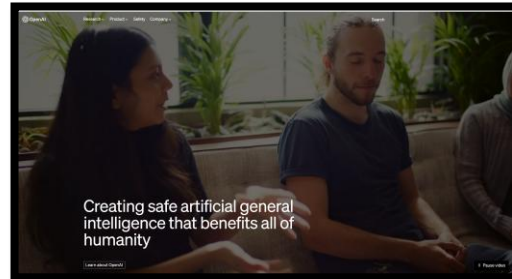


# Social Media Ads in Video Domain -2

- There are **many types** of online video Ads on social media platforms.
- **Roll Ads** refers to the paid, promotional short videos that play with featured videos. Roll Ads can be embedded before, after, or even in the middle of the main videos. Some roll Ads are typically skippable and can be up to 30 seconds long.
- **In-banner video ads** are short videos (with no sound) or GIFs embedded into a typical display (or banner) ad slot.



Roll videos on YouTube



In-banner video ad on OpenAI's main page

# Social Media Ads in Video Domain -3

- **E-commerce streams** are one of the representative advertising products on influencer marketing.
  - Companies can pay **influencers** on live stream platforms to promote and advertise their products.
  - Recently, on online shopping platforms like **Taobao**, some companies have also begun advertising on live stream by themselves, by which they could increase the interaction and engagement with customers.
  - E-commerce streams have become viral in Chinese markets. It is reported by Taobao that, in 2021, there are more than 460 million users watching **e-commerce Ads on live streams** (Source).
  - On the Double 11 in 2024, the sales performance of e-commerce via livestream was over 332 billion (Source).



# Unhealthy Food Marketing to Children - 1

- However, sometimes Social Media Ads can go wrong. One of the common issues is **the exposure of children to unhealthy Food Ads**.
- For food advertising, it is widely recognized that most Ads content focus particularly on **unhealthy and palatable food products**.
  - For instance, it is reported that, in U.S., a stunning 98% of advertised food products are considered unhealthy by nutritionists (Powell, 2007), typically being energy-dense and high in added sugar, salt, and fat.
- As for children marketing, systematic research reviews have repeatedly shown that food advertising **affects children's eating behavior**, contributing to **childhood obesity** (Harris, 2009).
  - Researchers have shown significant evidence that children exposed to **junk food Ads** are more likely to have higher and unhealthy BMI scores, suggesting a high risk of obesity for these children (Andreyeva, 2011).

# Unhealthy Food Marketing to Children - 1

- The issue of unhealthy food Ads has also been observed on social media platforms.
- Results from related study suggest that young YouTube influencers are increasingly marketing junk food to fellow kids (CNN).
- Researchers analyzed 418 videos posted by the five most-watched kid influencers (age 3~14) on YouTube in 2019 and found that 179 of the videos featured food or drinks.
  - With 90% of those instances showing unhealthy branded items, such as fast food.

*This video is an example of junk food videos by Ryan's World, a young YouTube influencer with 34+ million followers. This video records how Ryan DIY a hamburger with snacks by McDonalds' Hamburger Maker, the toy products Ryan advertises ([YouTube](#)).*



# Unhealthy Food Marketing to Children - 3

- To deal with such problems, many social media platforms have released regulations to posted Ads content, in order to **decrease children's exposure to non-suitable Ads**.
  - i.e., YouTube requires YouTubers to check a box to inform whether there are sponsored content non-suitable for kids before uploading their videos, by which the platform will filter these Ads for young viewers.
- However, **companies can still circumvent these regulations in some virtual communities**. For examples, **alcohol advertisers** can create access to their brand pages throw user profiles, throw which young users still receive Ads.
- Barry et al. took a closer look on these issues. By monitoring teenager profiles on Twitter (Now X) and Instagram (N=20), their findings pointed out that **no restriction to viewing and interacting with Ads on alcohol brand pages on these platforms**, where these users had received **362 Ads with in 30 days** (Barry, 2016).

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  - i.e., YouTube requires YouTubers to check a box to inform whether there are sponsored content non-suitable for kids before uploading their videos, by which the platform will filter these Ads for young viewers.
- However, **companies can still circumvent these regulations in some virtual communities**. For examples, **alcohol advertisers** can create access to their brand pages throw user profiles, throw which young users still receive Ads.
- Barry et al. took a closer look on these issues. By monitoring teenager profiles on Twitter (Now X) and Instagram (N=20), their findings pointed out that **no restriction to viewing and interacting with Ads on alcohol brand pages on these platforms**, where these users had received **362 Ads with in 30 days** (Barry, 2016).

# Ads and **Augmented Reality** - 1

- With the rise of immersive technology, new forms of advertising have emerged, and one of the most exciting is augmented reality (AR).
- AR is a technology that overlays digital content, such as **images, videos, and 3D models**, onto the real world.
  - AR can be experienced through a smartphone or tablet, using a camera and screen, or through specialized headsets (E.g., HoloLens 1&2).
- AR advertising allows businesses to create engaging and immersive experiences for customers.
  - For instance, IKEA Place is a mobile app can use AR to allow customers to see how a piece of furniture would look in their home before making a purchase.



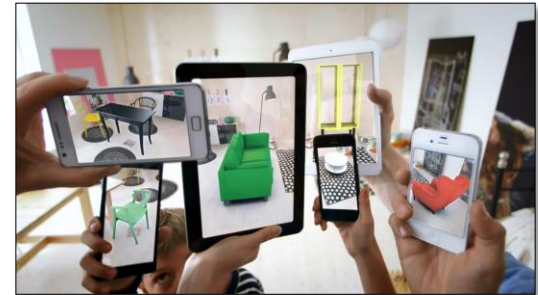
# Ads and **Augmented Reality** - 2

- AR advertising can also be **interactive**, allowing customers to engage with the brand in a new way.
- For instance, **BMW** launches AR Experience in 2022. BMW's AR experience allows customers to digitally place BMW i4 or iX into any real-world environment. Moreover, users can explore the features of a new car, customize the color and options, and even take a virtual test drive (Source).
- AR advertising can also be more **effective** than traditional advertising as it creates a memorable experience that engages customers on a deeper level.
- It can also be more **cost-effective** because it doesn't require physical products or spaces for customers to interact with.



# Ads and **Augmented Reality** - 3

- However, AR advertising are still facing several **challenges** on supporting technologies.
- Ruyter et al. performed a systematic review on AR advertising's challenges and opportunities. The authors summarized **several key technologies** for building AR Ads apps (Ruyter, 2020).
- **Image classification**
  - As technology develops, computer vision can further classify physical objects to identify and interpret what they are in AR displays. This is done by incorporating aspects of machine learning with the image data scanned by an AR device.
  - For example, the **IKEA Place app** to recommend matching furniture items to those it recognizes in a customer's room and subsequently place them into the view of the physical environment as lifelike 3D holograms.



# Ads and **Augmented Reality** - 4

## ● **Real-time Analytics**

- As technology develops, computer vision can further classify physical objects to identify and interpret what they are in AR displays. This is done by incorporating aspects of machine learning with the image data scanned by an AR device.
- For example, the **IKEA Place app** to recommend matching furniture items to those it recognizes in a customer's room and subsequently place them into the view of the physical environment as lifelike 3D holograms.

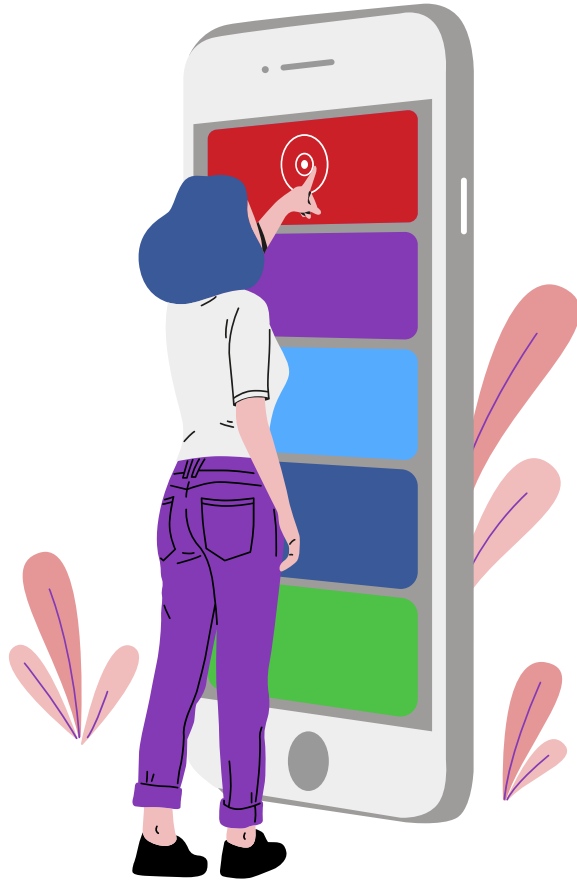


## ● **Gaze Tracking**

- By tracking a customer's gaze, an **advertiser** also knows what the customer is currently looking at.
- This information helps advertisers to estimate a customer's perspective toward objects and analyze **a pattern of gaze fixations**.



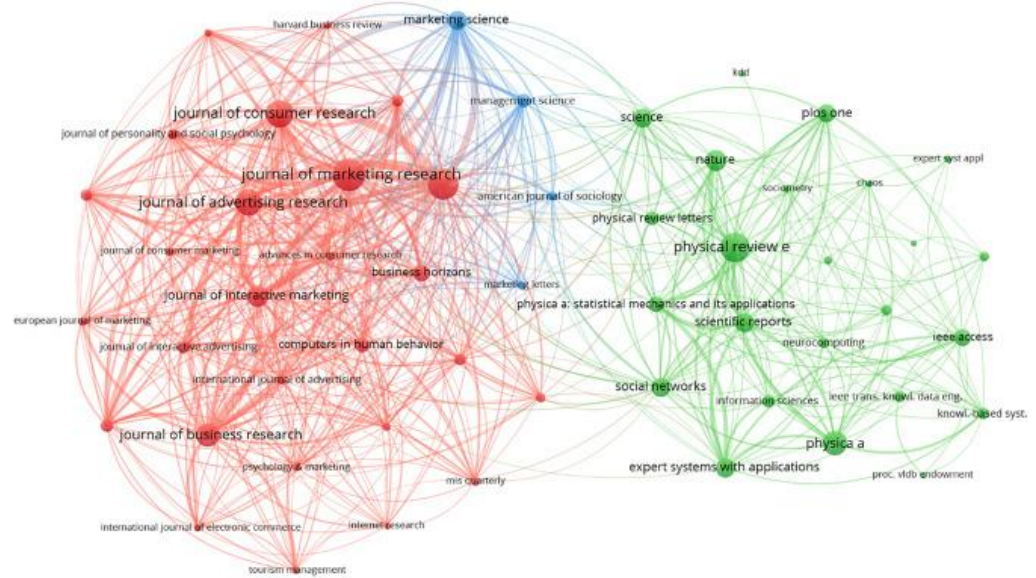
# Today's Outline



03

## Lecturing

Homophily, Creative Advertisement and Viral Marketing



VOSviewer

# Viral Marketing

- **Viral marketing** refers to a diffusion of information about the product and its adoption over the network.
- In the context of **social media**, viral marketing exploits existing social networks by encouraging customers to share product information with their followers or friends, thus creating a "**viral**" effect.
- The **key** to **successful** viral marketing on social media is to understand customers' social networks.
  - It has been shown that **social network properties** (e.g., social influence, homophily, ties strength, etc.) can play significant role on **content propagation in viral marketing**, by determining users whether would purchase or share the products (Leskovec, 2007).



## VIRAL MARKETING



# Hotmail : Success of Viral Marketing - 1

- One widely cited success story of viral marketing on social media is **Hotmail.com** free email address campaign.
- When Hotmail launched in 1996, the internet was still a relatively new phenomenon, and **web-based email** was practically unheard of.
- Hotmail's founders, **Sabeer Bhatia** and **Jack Smith**, recognized the potential of email as a viral marketing tool.
- They realized that if they could get people to sign up for Hotmail, those people would automatically promote the service by including a "**Get your free email at Hotmail**" tagline at the bottom of each email they sent. → Slogan!!!

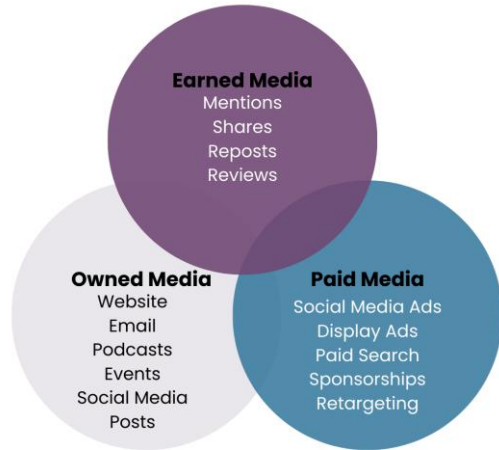


# Hotmail : Success of Viral Marketing - 2

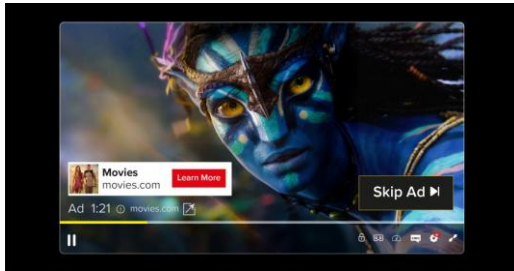
- To get the ball rolling, Hotmail launched with **a simple, yet effective, marketing message**: "Get Your Free Email at Hotmail."
- The message spread quickly, and Hotmail's user base grew rapidly. Within six months of its launch, Hotmail had over **one million users**.
- By 1997, Hotmail had attracted over 8 million users, and it was acquired by Microsoft for a reported \$400 million.
- As pointed out by researchers, Hotmail's success as viral marketing can be attributed to that
  - *"It makes users become **automatically 'salesmen' of the firm** as Hotmail's ad-financed service package and its own advertisers' messages are promoted by every user's email-notes."* (Helm, 2010)



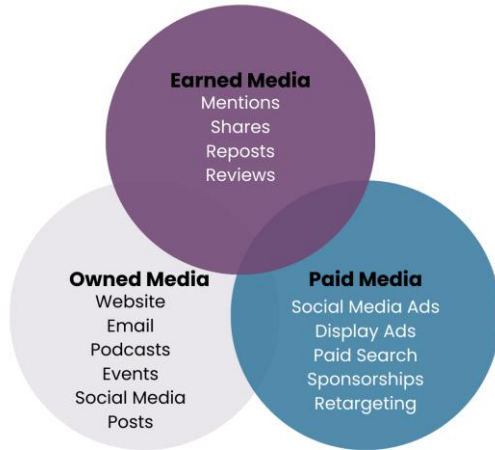
# Paid, Owned, and Earned Media - 1



- In the context of viral marketing, marketers have adopted a typology for offline and online media activity into three categories: **paid media**, **owned media**, and **earned media**.
- **Paid media** refers to any type of advertising or promotional content that is paid for by the marketer.
  - This can include **social media ads**, **search engine ads**, **display ads**, and **influencer marketing**.
  - Paid media allows marketers to **target specific audiences and reach a large number of people quickly**, which contributes to generating initial interest in a viral campaign and drive traffic to owned media channels.
  - i.e., **rolled video Ads on YouTube**

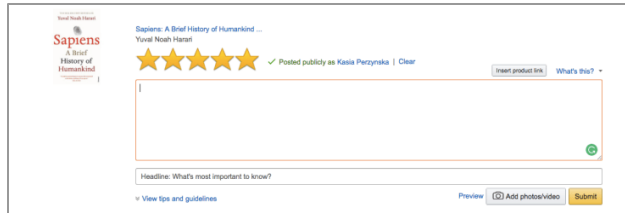
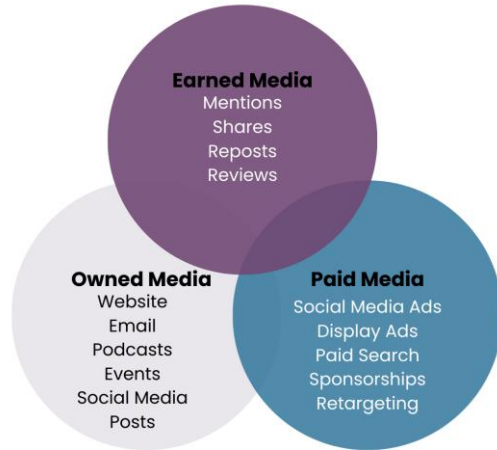


# Paid, Owned, and Earned Media - 2



- **Owned media** refers to any content or media that is owned and controlled by the marketer.
  - This can include a **companies' website, official profiles pages, social media accounts, and email newsletters.**
  - Owned media provides a **platform** for the marketer to share content and engage with their audience.
  - Owned media also allows marketers to **control** the messaging and branding of their campaign, which can be critical in building **trust and credibility** with consumers.
  - i.e., companies' **official e-commerce live channels on Taobao.**

# Paid, Owned, and Earned Media - 3



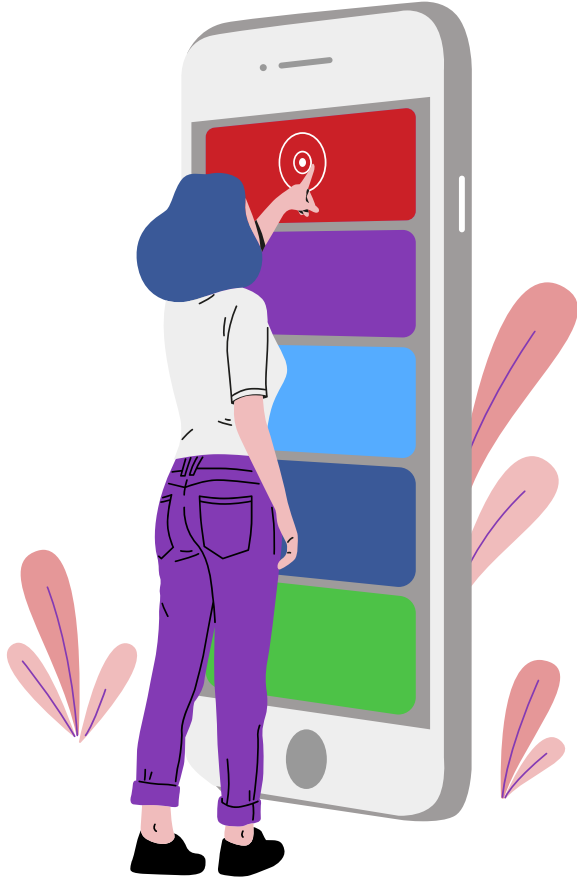
- **Earned media** refers to any content or media that is created by consumers or third-party sources.
  - This can include social customers comments, blog reviews, product reports articles, and word-of-mouth recommendations.
  - Most earned media is based on **social proof** and **authentic endorsements** from real people.
  - Earned media is also **highly shareable** and can **spread** quickly across social networks, amplifying the reach and impact of a viral campaign.
  - i.e., **Amazon customer reviews page**.

# Today's Outline

04

## Paper Reading and Presentation

Group 1: Social Media for Academia

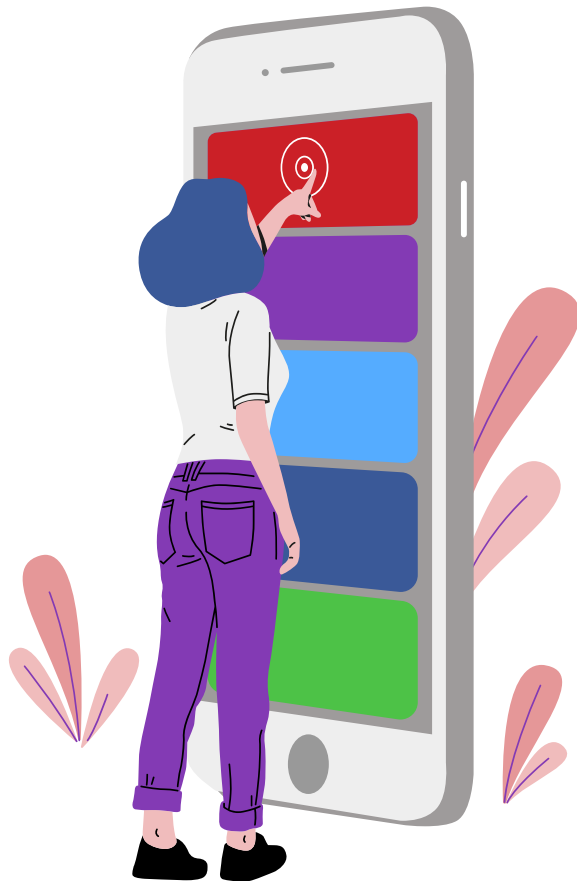


# Today's Outline

05

## Participation and Group Discussion

Template



Note! If you find any question(s) irrelevant/unsuitable for your chosen phenomenon, please feel free to change and note it in text color red.  
DO NOT EDIT ON THIS PAGE, YOU CAN ONLY COPY AND PASTE THE INFORMATION AS THIS IS A SHARED TEMPLATE.

### Group 0

FrameVR URL: Put your link of FrameVR here  
Group members: Student One Name (SID: 1234 5678: CWB/GZ/UH)  
Student Two Name (SID: 1234 5678: CWB/GZ/UH)  
Student Three Name (SID: 1234 5678: CWB/GZ/UH)  
Student Four Name (SID: 1234 5678: CWB/GZ/UH)

#### Topic / Chosen Phenomenon:

| The Power of Multimedia in Social Media Platforms                                       | Heterogeneity, Creative Advertising, and Viral Marketing                                | Social Networks and Influence                                                            | Digital Storytelling and Citizen Journalism                                              |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Week 4</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br> | <b>Week 5</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br> | <b>Week 6</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br>  | <b>Week 7</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br>  |
| <b>Week 8</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br> | <b>Week 9</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br> | <b>Week 10</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br> | <b>Week 11</b><br>Task 1: Individual<br><br>Task 2: VR Group<br><br>Task 3: After VR<br> |

Follow the announcement for more info.  
Template:

[https://miro.com/app/board/uXjVG4ZqvCU=?share\\_link\\_id=774962572902](https://miro.com/app/board/uXjVG4ZqvCU=?share_link_id=774962572902)

# Today's Outline

05

## Participation and Group Discussion

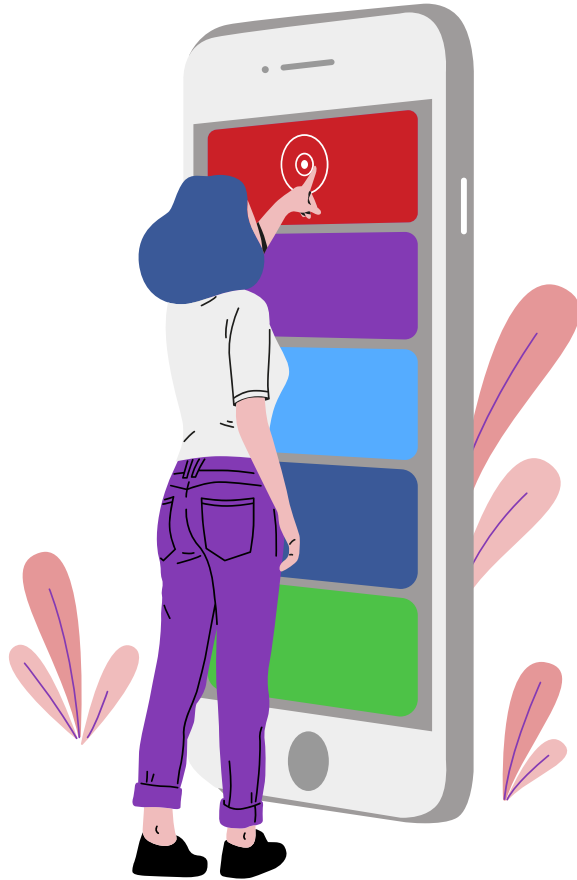
### Guidelines

| Group Number | Group Member<br>(* indicates the group leader) | Campus (CWB/GZ/Helsinki)                              | eMail                             | Your Phenomenon (by Week 4)                                  | Group Miro Board Link<br>(by Week 4)                                    | FrameVR<br>(https://framevr.io/)                                        | Week for Paper Reading<br>Presentation (from Week<br>5-9) |
|--------------|------------------------------------------------|-------------------------------------------------------|-----------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------|
| 1            | Ka Man (Janet) CHOI                            | CWB (PhD student in IIP - XR and wearables)           | kmchoi546@connect.ust.hk          | social media for academia                                    | <a href="https://framevr.io/emi6500-c">https://framevr.io/emi6500-c</a> | <a href="https://framevr.io/emi6500-c">https://framevr.io/emi6500-c</a> | 5                                                         |
|              | Melling WEI                                    | CWB (PhD student in IIP - AI and media communication) | mweiai@connect.ust.hk             |                                                              |                                                                         |                                                                         |                                                           |
|              | Yuqi LIANG                                     | GZ (PhD student in CMA)                               | ylang379@connect.hkust-gz.edu.cn  |                                                              |                                                                         |                                                                         |                                                           |
|              | Tianzuo SHI                                    | GZ (PhD student in CMA)                               | tsu886@connect.hkust-gz.edu.cn    |                                                              |                                                                         |                                                                         |                                                           |
| 2            | Ruoshan (Bianca) Yang                          | GZ (PhD student in CMA - Digital Cultural Heritage)   | ryang565@connect.hkust-gz.edu.cn  | The Sound of Self: Identity and Community in Music Streaming | <a href="https://framevr.io/group2-c">https://framevr.io/group2-c</a>   | <a href="https://framevr.io/group2-c">https://framevr.io/group2-c</a>   | 6                                                         |
|              | Tianhui LIU                                    | GZ (PhD student in CMA)                               | liu883@connect.hkust-gz.edu.cn    |                                                              |                                                                         |                                                                         |                                                           |
|              | Runqian (Spencer) Yang                         | GZ (MPhil student in CMA)                             | spencery@connect.hkust-gz.edu.cn  |                                                              |                                                                         |                                                                         |                                                           |
|              | Samar Mahajan                                  | Helsinki                                              | samar.mahajan@helsinki.fi         |                                                              |                                                                         |                                                                         |                                                           |
| 3            | Yuelu LI                                       | GZ                                                    | lyl883@connect.hkust-gz.edu.cn    |                                                              | <a href="https://framevr.io/group3-5">https://framevr.io/group3-5</a>   | <a href="https://framevr.io/group3-5">https://framevr.io/group3-5</a>   | 7                                                         |
|              | Xinyi Chen                                     | GZ                                                    | xcchen822@connect.hkust-gz.edu.cn |                                                              |                                                                         |                                                                         |                                                           |
|              | Yulin Yao                                      | GZ                                                    | yyao294@connect.hkust-gz.edu.cn   |                                                              |                                                                         |                                                                         |                                                           |
|              | Sam Lin                                        | CWB (Ms Financial Maths)                              | sam.lin@connect.ust.hk            |                                                              |                                                                         |                                                                         |                                                           |
| 4            | JIAN YANG                                      | GZ (PhD student in CMA - )                            | iyang000@connect.hkust-gz.edu.cn  | Social Media in Political Science                            | <a href="https://framevr.io/emi6500-c">https://framevr.io/emi6500-c</a> | <a href="https://framevr.io/emi6500-c">https://framevr.io/emi6500-c</a> | 8                                                         |
|              | Feiyue ZHAO                                    | GZ (PhD student in CMA)                               | fzhaoy322@connect.hkust-gz.edu.cn |                                                              |                                                                         |                                                                         |                                                           |
|              | YOU ZHOU (YOYO)                                | GZ (PhD student in CMA - Digital Cultural Heritage)   | yzhou785@connect.hkust-gz.edu.cn  |                                                              |                                                                         |                                                                         |                                                           |
|              | Zhang Jingqi (Carrie)                          | CWB (PhD student in IIP - AI and media communication) | jzhang1@connect.ust.hk            |                                                              |                                                                         |                                                                         |                                                           |
| 5            | Fabian Wangsa Saputra                          | GZ (MPhil student in CMA)                             | sws931@connect.hkust-gz.edu.cn    |                                                              | <a href="https://framevr.io/emi6500-c">https://framevr.io/emi6500-c</a> | <a href="https://framevr.io/emi6500-c">https://framevr.io/emi6500-c</a> | 9                                                         |
|              | Yixuan XIE (Evelyn)                            | CWB                                                   | yxex@connect.ust.hk               |                                                              |                                                                         |                                                                         |                                                           |
|              | Duorong Wang                                   | CWB (PhD student in IIP)                              | dwangb@connect.ust.hk             |                                                              |                                                                         |                                                                         |                                                           |

**Chosen  
Phenomenon / Topic  
that will be used in  
this class as your  
Paper reading topics  
+ Group discussion  
topics!**

**Miro Board and  
FrameVR**

# Today's Outline



05

## Participation and Group Discussion

### Task 1: Individual

Built on your topic/chosen Phenomenon. Look for commercial and viral dynamics. Add to Miro:

- Is there marketing happening around your phenomenon
- How does virality work here?
- Do you see homophily (like attracting like)?
- Who's profiting? One example of viral or commercial activity.

# Today's Outline



05

## Participation and Group Discussion

### Task 2: Group

Share ideas via FrameVR. Discuss:

- Do we agree on the best methods?
- What would quantitative vs qualitative approaches reveal?
- What's hard to study about this phenomenon?
- What ethical issues might arise?
- Identify one strong research question, one methodological disagreement.

# Today's Outline



05

## Participation and Group Discussion

Task 3: Group + Individual

Create "Week 5: Research Methods" section with:

- proposed research question, methods that would work, data sources available, limitations and ethics, connections to previous observations.
- Individual reflection: What would you most want to know about this phenomenon? Why is it hard to answer?

# Today's Outline

05

## Participation

Fill in the survey that counts your participation score

